

Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

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Abstract. Natural disasters severely disrupt transportation infrastructure, heavily complicating search and rescue (SAR) operations. Existing humanitarian logistics models often fail to capture these dynamic disruptions and the non-linear escalation of human suffering. To address this gap, we propose a two-stage stochastic Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP). Our model simultaneously minimizes expected logistics costs – utilizing Daganzo’s Continuous Approximation for scalable last-mile routing – and the expected maximum deprivation cost to enforce social equity. To solve this NP-hard problem, we develop a Priority-Based Non-dominated Sorting Genetic Algorithm II (PB-NSGA-II). Extensive computational experiments on standard benchmarks (AP, TR81) validate the algorithm’s efficiency. Furthermore, a detailed case study of the flood-prone Vu Gia – Thu Bon river basin provides crucial managerial insights. The results highlight the necessity of dynamically adapting multi-modal network topologies to maintain resilience under extreme disaster scenarios.

Keywords: Humanitarian Logistics · Disaster relief · Hub location problem · Multi-objective Evolutionary Algorithm.

1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [?, ?]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [?, ?].

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There were many great challenges in humanitarian operations during this period. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains [?], while key arteries such as National Highway 1 were severed by landslides, isolating communities for days [?]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes “incomplete”, characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster management involves four phases: mitigation, preparedness, response, and recovery [?]; Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing.

1.2 Related Works

Humanitarian logistics and facility location. Humanitarian logistics differs fundamentally from its commercial counterpart in that speed and equity dominate over cost minimisation [?, ?]. A comprehensive review of humanitarian facility location under uncertainty [?] identifies stochastic programming and robust optimisation as the dominant paradigms, while Boonmee et al. [?] document the growing need for models that jointly capture disruption, uncertainty, and equity. Critically, Holguín-Veras et al. [?, ?] argued that purely cost-based objectives neglect social welfare, and proposed *deprivation cost* – an exponential penalty on victim waiting time – as the appropriate humanitarian objective.

Hub location and incomplete networks. Hub-and-spoke structures consolidate flows efficiently and have been studied extensively since Campbell’s seminal formulations [?]. For disaster settings, the *incomplete* hub network – in which infrastructure damage severs inter-hub arcs – is a more realistic topology than the classical fully-connected graph [?]. Sangsawang and Chanta [?] proposed a pioneering flood-specific hub location model, a single-objective deterministic formulation for Thailand, establishing that hub-and-spoke structures yield measurable gains in relief distribution efficiency. Multi-modal hub networks further address the heterogeneous transport modes (road, waterway, air) that characterise disaster response [?].

Stochastic and robust optimisation for relief networks. The two-stage stochastic programming framework naturally separates pre-disaster strategic decisions from post-disaster recourse [?, ?]. Under demand and supply uncertainty, robust formulations [?, ?] and prepositioning models [?] have been shown to outperform deterministic baselines. Disruption of transportation infrastructure compounds this uncertainty by rendering pre-planned routes infeasible [?], a condition endemic to Central Vietnam’s flood disasters.

Multi-objective evolutionary algorithms. NSGA-II [?] is the standard framework for multi-objective combinatorial problems in humanitarian logistics, owing to its parameter-free Pareto ranking and elitist selection. Zhou et al. [?] applied a multi-objective evolutionary algorithm to multi-period dynamic

emergency resource scheduling under uncertain road networks, demonstrating the effectiveness of population-based search when scenario-dependent decisions are coupled to shared strategic variables. Li et al. [?] designed a bi-objective multimodal hub-and-spoke network for emergency relief using a meta-heuristic approach, showing that problem-specific customized operators are essential for maintaining feasibility across complex network topologies.

Research gap and contributions. No existing work simultaneously addresses incomplete hub network design, two-stage stochastic programming, multi-objective humanitarian objectives (logistics cost and deprivation cost), and a flood-disaster case study. Sangsawang and Chanta [?] is the closest precedent, yet it is deterministic and single-objective. The present paper closes this gap by proposing MO-IHLNDP – a two-stage stochastic multi-objective incomplete hub location model integrating Daganzo’s CA for last-mile cost estimation – and PB-NSGA-II, a priority-based [?] evolutionary algorithm that efficiently reconstructs second-stage decisions across flood scenarios in Central Coastal Vietnam.

2 Problem Description and Mathematical Model

To design an efficient disaster relief network (DRND) for flood disaster response in Central Vietnam, we frame the problem as a multi-objective incomplete hub location network design problem (MO-IHLNDP) – the capacitated, single-allocation variant of the HLP. The network comprises three main components: origins, hubs, and demands. Hubs refer to rescue stations of two kinds: the first is established in advance of the disaster, strategically positioned and pre-inventoried with amenities; the second is opened dynamically during response, prioritizing approachability to affected areas. **Demands** refer to victim groups discovered progressively during operations, while **origins** represent both government-planned forces and spontaneous volunteers.

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption – damaged roads and buildings – can disable pre-programmed plans, urging the need for a dynamic solution [?]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [?]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that SAR operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

Finally, the model places strong emphasis on *social equity* by integrating the deprivation cost [?], and adopts Daganzo’s continuous approximation [?] to obtain a tractable yet robust estimate of rescue time from each hub to its assigned demand points.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uv}, τ_{uv}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F_{ks}^a	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvms}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u , and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i , and relief items provided at origin j in scenario s
c_{jks}	Pre-computed cheapest transport cost from origin j to hub k in scenario s : $c_{jks} = \min_{m \in \mathcal{M} a_{jkm.s}=1} C_{jkm}$, ($+\infty$ if no available mode)
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuitry factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s , 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s , 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s , 0 otherwise
$q_k \geq 0$	Amount of relief items inventoried at hub k
$f_{khms} \geq 0$	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k \in \mathcal{H}} F_k x_k + \sum_{k \in \mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s \in \mathcal{S}} \pi_s \left[\underbrace{\sum_{k \in \mathcal{H}} F_{ks}^a y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{H}} c_{jks} \cdot O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \right. \\
& \left. + \underbrace{\sum_{k \in \mathcal{H}} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k \in \mathcal{H}} \sum_{i \in \mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \right] \quad (1)
\end{aligned}$$

Objective Z_1 minimizes the total expected logistics cost, balancing pre-disaster investments against reactive setups and transportation costs. To have an accurate estimation of last-mile evacuation cost while keeping the model simple, we adopted Daganzo's Continuous Approximation [?]. The region of interest is partitioned via spatial discretization, and the demands are aggregated on a grid basis. The routing cost Θ_{kis} is pre-computed as a parameter:

$$\Theta_{kis} = \min_{m \in \mathcal{M}} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s \in \mathcal{S}} \pi_s \left(\max_{i \in \mathcal{I}} [D_{is} \cdot (e^{\lambda_{is} \cdot \Omega_{is}} - 1)] \right) \quad (2)$$

The second objective ensures social equity by minimizing the expected maximum deprivation cost across all scenarios. Based on Holguín-Veras et al. [?], Z_2 adopts an exponential function to penalize long waiting times (Ω_{is}) for victims. Furthermore, it forces the model to distribute resources evenly, ensuring overall low delay time for everyone.

$$\Omega_{is} = \sum_{k \in \mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M}} \tau_{kim} \right)$$

The sensitivity coefficient λ_{is} is proportional to the risk index r_{is} ($\lambda_{is} = \lambda_0(1 + r_{is})$), effectively prioritizing vulnerable populations. Although Z_2 introduces non-linearity, our meta-heuristic solution approach can directly evaluate it as a fitness function. For linearization into a Mixed-Integer Linear Programming (MILP) form, one can exploit the single-allocation property ($\sum_{k \in \mathcal{H}} z_{iks} = 1$).

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$\text{Minimize } Z_1, Z_2$$

$$\text{subject to Constraints (??) – (??)}$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (3)$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \quad (4)$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (5)$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \quad (6)$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \quad (7)$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \quad (8)$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \quad (9)$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \quad (10)$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \quad (11)$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \quad (12)$$

$$\begin{aligned} \sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} &\leq q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} \\ &+ \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \quad (13)$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (14)$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \quad (15)$$

$$q_k, f_{khms} \geq 0 \quad \forall k, h, m, s \quad (16)$$

Constraints (??) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (??), while the reactive hubs are required to be placed within safe zones (??). Constraints (??)-(??) rigorously enforce the single-allocation property to active hubs via available links. Constraint (??) restricts flow to be trans-shipped on intact paths. Constraints (??) and (??) maintain flow conservation and capacity limits. Specifically, (??) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (??)-(??).

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [?]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [?] with a custom heuristic decoder.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0, 1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0, 1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0, \dots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i . The continuous vector $\mathbf{W} \in [0, 1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency ($\lambda \cdot D$) and isolation; w_1 , w_2 , w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$; if none qualify, the safest hub is forced active. The proximity order π_k , ranking all hubs by ascending distance from anchor hub k , is pre-computed once.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score

$$\text{Score}_i = w_0 \widetilde{\text{urgency}}_i + w_3 \widetilde{\text{isolation}}_i - w_1 \widetilde{\text{dist}}_i + \varepsilon_i, \quad (17)$$

where $\text{urgency} = \lambda_{is} D_{is}$, $\text{isolation} = 1/|\mathcal{H}_{\text{act},i}|$, and dist = minimum travel time to any active hub, each normalised to $[0, 1]$; $\varepsilon_i \sim \mathcal{N}(0, \sigma)$ prevents premature convergence. Demands are processed in descending score order.

Step 3 – Tiered Hub Assignment. For each demand i , the trial order follows π_{A_i} . Let $K = \max(1, \lceil w_5 \cdot |\mathcal{H}| \rceil)$. *Pass 1* selects the hub maximising $\text{HubScore}(k) = w_1/\tau_{ikm^*} + w_2 \cdot \text{residual}_k + w_4 \cdot \mathbf{1}[X_k = 1]$ among the K closest-to-anchor hubs that are active, reachable, and have positive residual inventory. If *Pass 1* fails, *Pass 2* assigns i to the first active reachable hub from the remaining candidates (capacity violation permitted). If both passes fail, a *reactive fallback* opens the nearest safe inactive hub at cost F_{ks}^a ; irrecoverable infeasibility accumulates as constraint violation.

Step 4 – Supply Routing and Objective Accumulation. Origins are greedily assigned to the most under-supplied active hub. Surplus inventory is redistributed via inter-hub trans-shipment with discount α . Z_1 and Z_2 are accumulated as probability-weighted expectations over all $s \in \mathcal{S}$. Algorithm ?? formalises Steps 1–4.

Algorithm 1 Priority-Based Decoder (single scenario s)

Require: Chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$, scenario s , instance data

Ensure: (Z_1^s, Z_2^s, CV^s)

```
1: Activate  $\mathcal{H}_{\text{act}} \leftarrow \{k : X_k = 1, r_{ks} \leq \chi\}$ ; set  $q_k \leftarrow R_k \kappa_k$ 
2: if  $\mathcal{H}_{\text{act}} = \emptyset$  then
3:   Force-activate  $\arg \min_k r_{ks}$  among  $\{k : X_k = 1\}$ 
4: end if
5: Score each demand  $i$  by Eq. (??); sort in descending order
6: for each demand  $i$  (in priority order) do
7:    $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$ ; trial order  $\leftarrow \pi_{A_i}$ 
8:   Pass 1:  $k^* \leftarrow \arg \max \text{HubScore}(k)$  over first  $K$  active, reachable hubs with residual  $> 0$ 
9:   if Pass 1 succeeds then
10:    Assign  $i \rightarrow k^*$ ; deduct demand from residual inventory of  $k^*$ 
11:   else if any active reachable hub remains then
12:    Pass 2: assign  $i$  to first such hub (capacity violation allowed)
13:   else
14:    Reactive fallback: open nearest safe inactive hub; add  $F_{ks}^a$ ;  $CV^s += 1$ 
15:   end if
16: end for
17: Assign origins greedily; redistribute surplus via inter-hub trans-shipment (discount  $\alpha$ )
18: Accumulate  $Z_1^s, Z_2^s$ 
19: return  $(Z_1^s, Z_2^s, CV^s) = 0$ 
```

3.3 Genetic Operators and Selection

Selection uses binary tournament on Pareto rank with crowding distance as tiebreaker. *Crossover* ($p_c = 0.9$) applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and SBX ($\eta_c = 20$) to $\mathbf{R}$ and $\mathbf{W}$. *Mutation* ($p_m = 1/|\mathbf{C}|$) applies Bit-flip to $\mathbf{X}$, random-reset to $\mathbf{A}$, and Polynomial mutation ($\eta_m = 20$) to $\mathbf{R}$ and $\mathbf{W}$. Constraint handling follows the constrained-dominance principle [?]: feasible solutions always dominate infeasible ones; among infeasible solutions, lower CV dominates. When a partial Pareto front must be split, tiebreaking proceeds by (1) Pareto rank, (2) crowding distance, (3) minimum Hamming distance in $\mathbf{X}$ -space – preserving genotypic diversity in hub configuration and countering premature convergence.

4 My draft – experiments

4.1 Experimental Setup

- **Programming language and machine configuration.** Plan to use the DIMACS 2011 benchmark for machine speed measurement, if space allows.
- Algorithm parameters, and parameters for problem instances.
- Evaluation metrics. Cite Hypervolume and IGD, and explain briefly in just 1-2 sentences for each

- For statistical stability, each experiment is run for 20 times. Then the mean and standard deviation of metric values are recorded. Each run use a consistent random seed per iteration.

4.2 Dataset Description

Benchmark dataset. As no benchmark dataset is available for the MO-IHLNDP, we transformed HLP benchmarks. We adapt two widely-used hub location benchmarks: the **AP** set [?] (Euclidean instances of sizes 10–100 nodes) and **TR81** [?] (Turkish inter-city matrix, 81 nodes). Have a very brief paragraph that summarize the data calibration process (please based on our `process_benchmark.pyfile`)

Synthetic dataset for Central Region Several coastal cities and provinces at the Central Vietnam, namely Da Nang, Hue, Quang Tri and Quang Ngai are considered.

- Coordinates collection
- Building auxiliary risk matrix
- Constructing distance and travel time matrix
- Scenarios generation (epicenter strategy)
- Large instance and Small instance
- What's more?

4.3 Experiment 1: Algorithms benchmarking

- Dataset: Standard benchmark dataset (HLP -i DRND)
- Compare the proposed algorithm against complete enumeration technique (speed up with branch and bound solver). Prove that the PG-NSGA produces optimal results for small instance
- Stress-testing (measuring running time in seconds) and showcasing algorithm's efficiency

4.4 Experiment 2: Case study in Vietnam

- Dataset: Our synthetic dataset. Only run the proposed algorithm on the dataset.
- Sensitivity analysis. How does the solution structure change between different scenarios.
- Managerial insights

The figures that need to be inserted:

- Pareto front (used for trade-off analysis)
- Solutions found by the algorithm, plotted on the Vietnamese map. What are the implications and managerial insights?
- Do a slight sensitivity analysis from the same figure (solution visualized on map)

5 Computational Experiments

5.1 Experimental Setup

Implementation. PB-NSGA-II is implemented in C++17. Data generation and result analysis are performed in Python 3. All experiments run on a standard workstation (Intel Core i7, 16 GB RAM).

Algorithm parameters. Population $N = 100$; generations $G = 200$; $p_c = 0.9$; $\eta_c = \eta_m = 20$; $p_m = 1/|\mathbf{C}|$. The safety threshold is $\chi = 0.7$; discount $\alpha = 0.6$; $\gamma = 3.0$ kg/person; $\lambda_0 = 0.8$.

Performance metric. We use the **Hypervolume (HV)** indicator. The reference point is set to $1.1 \times \max(Z_1, Z_2)$ of the combined Pareto front. A larger HV indicates better coverage and spread of the Pareto front.

5.2 Dataset Description

Benchmark datasets. We adapt two widely-used hub location benchmarks: the **AP** set [?] (Euclidean instances of sizes 10–100 nodes) and **TR81** [?] (Turkish inter-city matrix, 81 nodes). Each benchmark is transformed into a DRND instance. We assign hub/origin/demand roles and generate three flood scenarios (mild, severe, extreme) with probabilities $(\pi_1, \pi_2, \pi_3) = (0.60, 0.30, 0.10)$. Supply and demand are calibrated to $1.5\text{--}2.5\times$ total demand coverage. Road distances are normalized to a 500 km scale. Three transport modes operate concurrently (Table ??).

Table 1. Transport mode parameters.

Mode	Vehicle	Speed (km/h)	Cost (\$/km)	Capacity (pers.)	Disruption
0	Truck	35	2.0	60	Road-dependent
1	Motorboat	25	5.0	25	Low (flood-resilient)
2	Helicopter	150	40.0	10	None (always available)

Central Vietnam case study. We generate two synthetic instances modelling the flood-prone **Vu Gia – Thu Bon** river basin (Da Nang – Quang Nam region):

- **CV-Small:** $|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$ – small-scale planning instance.
- **CV-Large:** $|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$ – high-resolution regional instance.

Node coordinates are drawn from real geographic locations. Road travel times are estimated via the OSRM API with a terrain-adjusted cost factor for inland areas. Scenario epicenters are sampled from flood-prone sub-regions. Risk indices decay with distance from the epicenter. Hub capacities are calibrated post-scenario generation to ensure structural feasibility ($\kappa_k \geq 3\hat{D}/|\mathcal{H}|$).

5.3 Results

Table ?? reports the final-generation hypervolume for each instance. For benchmark instances, we run one seed ($G = 200$, $N = 100$). For CV-Small, we run three independent seeds to assess robustness.

Table 2. Hypervolume (HV) results across all instances. CV-Small reports mean over 3 seeds.

Category	Instance	#Pareto pts	HV
AP Benchmark	AP10	100	1.17×10^{14}
	AP20	100	6.77×10^{14}
	AP25	100	2.20×10^{13}
	AP40	100	9.87×10^{13}
	AP50	100	4.94×10^{13}
	AP100	25	2.65×10^{14}
TR Benchmark	TR81	2	3.20×10^{13}
Case Study	CV-Small ($n = 3$)	152 (combined)	1.01×10^{12}
	CV-Large	100	3.23×10^{13}

Benchmark observations. The algorithm consistently produces well-spread Pareto fronts across all AP instances. AP20 yields the largest HV. This reflects the favourable trade-off structure of mid-size instances. AP100 produces fewer non-dominated points (25). This is expected as the search landscape becomes harder and population diversity is consumed. TR81 converges to only 2 non-dominated solutions. It indicates a strong dominance structure where a single hub configuration dominates most others in a dense graph.

Case study observations. For CV-Small, a combined Pareto front of 152 points demonstrates excellent solution diversity and reproducibility. CV-Large achieves 100 non-dominated solutions with an HV comparable to TR81. This illustrates that PB-NSGA-II scales effectively to large real-world instances. The smaller HV relative to AP benchmarks is attributable to the narrow feasible region. The flood disruption model severely restricts options (e.g., only 3 out of 5 hubs remain safe in extreme scenarios).

5.4 Managerial Insights

Analyzing the Pareto front of CV-Small reveals a clear convex trade-off. Reducing Z_2 (deprivation) by 30% requires increasing Z_1 (logistics cost) by approximately $3\times$. This reflects the exponential structure of the deprivation function. Reducing waiting times for high-risk isolated areas is disproportionately costly.

Network topology analysis further highlights structural shifts. Under the *mild* scenario, the optimal strategy uses road transport from all 5 candidate hubs with moderate inventory pre-positioning. Under the *extreme* scenario, 2 hubs become

unsafe ($r_{ks} > \chi$). The network dynamically switches to water and air transport. The remaining active hubs receive larger inventory pre-commitments. This is compensated by increased lateral trans-shipment. This shift confirms the value of the two-stage stochastic formulation. A deterministic single-scenario model would systematically under-invest in network resilience.

6 Conclusion and Future Work

We proposed PB-NSGA-II for the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP). The model incorporates multi-modal transport, stochastic flood scenarios, deprivation cost, and Daganzo’s continuous approximation. Our priority-based decoding scheme efficiently handles second-stage variables and maintains feasibility in disrupted networks. Experiments on AP and TR81 benchmarks validate the algorithm’s effectiveness. The Central Vietnam case study demonstrates its practical applicability.

Limitations. The current model relies on a fixed set of pre-defined scenarios. The decoding heuristic is greedy and may miss globally optimal demand assignments.

Future work will explore exact branch-and-cut approaches for small instances to establish lower bounds. We also aim to incorporate real-time sensor data via online re-optimization and extend the framework to multi-period relief operations.

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